6G Zero-Drop: Neuro-Fuzzy Predictive Routing for Seamless LEO-Edge Handovers

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***Abstract*—Next-generation 6G networks envision seamless integration of terrestrial infrastructure with Low Earth Orbit (LEO) satellite constellations to achieve ubiquitous global coverage. However, the high orbital velocity of LEO satellites (approximately 7.5 km/s) results in frequent inter-satellite handovers, traditionally managed via reactive "Break-Before-Make" protocols. These protocols induce severe latency penalties of up to 3500 ms due to sequential execution of 6G-AKA' authentication, DHCPv6 IP reallocation, and BGP route convergence. This paper proposes a novel *Make-Before-Break* handover architecture that couples a Long Short-Term Memory (LSTM) neural network for time-series RSRP forecasting with a Mamdani-type Fuzzy Logic Inference System (FIS) for handover decision-making. The LSTM predicts signal degradation 3 seconds into the future, while the FIS evaluates prediction confidence and signal trend through seven linguistically interpretable rules to produce a continuous handover urgency score. When urgency exceeds the trigger threshold, 6G cryptographic protocols are initiated with the target satellite in the background while primary data continues to flow through the degrading link. Simulation results over 1000 seconds of physics-based orbital telemetry demonstrate a reduction in user-visible handover blackout from 3500 ms to 25 ms—a 99.3% improvement in network availability—while reducing total handover events from 19 to 1.**

***Index Terms*—6G Non-Terrestrial Networks, Low Earth Orbit, LSTM, Fuzzy Logic, Soft Computing, Predictive Handover, Make-Before-Break**

I. Introduction

The sixth generation (6G) of wireless communications is projected to unify terrestrial, aerial, and space-borne network segments into a single heterogeneous fabric [1]. Within this vision, Non-Terrestrial Networks (NTN) anchored by LEO satellite constellations (e.g., Starlink, OneWeb, Kuiper) are expected to deliver broadband connectivity to underserved regions, maritime vessels, and airborne platforms [2].

LEO satellites orbit at altitudes between 300–1200 km, completing a full orbit in approximately 90 minutes at velocities exceeding 7.5 km/s. This rapid motion implies that a ground user equipment (UE) device maintains line-of-sight with any single satellite for only 5–10 minutes before the satellite recedes below the radio horizon. The network must therefore execute frequent *inter-satellite handovers* to sustain continuous service.

A. The Handover Latency Problem

Conventional handover protocols operate reactively: the UE monitors the Reference Signal Received Power (RSRP) and triggers a handover only after the serving link degrades below a pre-defined threshold (typically -110 dBm). Upon triggering, the following 6G security and networking procedures must execute *sequentially*:

1) **EAP-AKA' Authentication** (1200 ms): Mutual authentication and session-key derivation between UE and target satellite.
2) **DHCPv6 IP Reallocation** (800 ms): Subnet-aware address reconfiguration and mobility binding update.
3) **BGP/ISL Route Convergence** (1500 ms): Inter-satellite link routing table propagation across the mesh backbone.

The cumulative latency of **3500 ms** constitutes a complete service blackout—unacceptable for mission-critical 6G use cases such as autonomous vehicle telemetry, remote surgery, and industrial IoT control loops.

B. Proposed Approach

This paper introduces a *Neuro-Fuzzy Predictive Router* that shifts handover management from a reactive to a *proactive* paradigm. The key insight is that RSRP degradation follows a deterministic trajectory governed by orbital mechanics, making it highly predictable. By forecasting RSRP 3 seconds before threshold crossing and initiating cryptographic protocols in the background, the user-visible interruption is reduced to the physical radio switching delay of 25 ms.

The contributions of this work are:

- A hybrid architecture combining LSTM-based time-series prediction with a Mamdani Fuzzy Inference System for interpretable handover decisions.
- A protocol-aware simulation that models the sequential 6G-AKA', DHCPv6, and BGP layers to justify latency improvements at the networking stack level.
- An interactive, open-source simulation dashboard deployed at https://6g-zero-drop-ai-satellite-handover-ia3kgnbtus29htksdpnnrp.streamlit.app/.

II. Related Work

Handover optimization in LEO satellite networks has been studied from multiple perspectives.

Threshold-based approaches. 3GPP Release 17 defines conditional handover (CHO) for NTN, where the UE evaluates RSRP against configurable thresholds with hysteresis

and time-to-trigger parameters [3]. While effective in slowly varying terrestrial cells, CHO struggles with the rapid signal dynamics of LEO passes, leading to either premature handovers or delayed triggers.

Machine learning methods. Recent works have applied reinforcement learning [4] and deep Q-networks [5] to optimize handover decisions in multi-beam satellite systems. However, these approaches treat the handover as a classification problem and do not address the underlying cryptographic latency.

Predictive handover. Several studies have employed Kalman filtering and linear regression to predict satellite visibility windows [6]. While computationally lightweight, these methods struggle with the non-linear RSRP dynamics caused by atmospheric noise and multi-path fading.

Fuzzy logic in handover. Fuzzy logic has been applied to vertical handover decisions in heterogeneous networks [7], leveraging its ability to handle uncertainty without requiring precise mathematical models. However, existing work does not combine fuzzy logic with deep learning prediction in the NTN context.

This paper uniquely bridges *deep learning prediction* with *fuzzy decision-making* and *protocol-layer simulation*, providing an end-to-end solution rather than an isolated algorithmic contribution.

III. System Model

A. Orbital Geometry and Signal Model

The simulation models two LEO satellites (Sat-A and Sat-B) orbiting at altitude $h = 500$ km above a stationary ground UE at position $x = 0$ on a 1D plane. Sat-A is initially directly overhead ($x_A(0) = 0$ km) while Sat-B trails at an offset of $x_B(0) = 1200$ km. Both satellites travel at orbital velocity $v = 7.5$ km/s.

The Euclidean slant distance between the UE and satellite $k \in \{A, B\}$ at time t is:

$$d_k(t) = \sqrt{(x_k(0) - v \cdot t)^2 + h^2} \tag{1}$$

The RSRP is derived from the Free Space Path Loss (FSPL) model with additive white Gaussian noise:

$$\text{RSRP}_k(t) = P_{tx} - \text{FSPL}(d_k, f) + \mathcal{N}(0, \sigma^2) \tag{2}$$

where $P_{tx} = 70$ dBm is the transmit EIRP, $f = 28$ GHz is the carrier frequency, $\sigma = 1.5$ dB models atmospheric scintillation, and:

$$\text{FSPL}(d, f) = 20 \log_{10}(d) + 20 \log_{10}(f) + 92.45 \ \ [\text{dB}] \tag{3}$$

A satellite is considered below the radio horizon when $|x_k(t)| > \sqrt{(R_E + h)^2 - R_E^2}$, where $R_E = 6371$ km is Earth's radius. Below-horizon RSRP is clamped to a floor of -130 dBm.

B. Simulation Parameters

Table I summarizes the simulation configuration. The environment generates 10,000 time-steps at $\Delta t = 0.1$ s, representing 1000 seconds of continuous orbital telemetry.

TABLE I
Simulation Parameters

Parameter	Value
Orbital altitude (h)	500 km
Satellite velocity (v)	7.5 km/s
Carrier frequency (f)	28 GHz
Transmit EIRP (P_{tx})	70 dBm
Atmospheric noise (σ)	1.5 dB
Simulation time-step (Δt)	0.1 s
Total time-steps (N)	10,000
RSRP handover threshold	-110 dBm
RSRP visibility floor	-130 dBm
Baseline processing delay	20 ms
Reactive handover penalty	3500 ms
AI soft-handover penalty	25 ms
Reactive cooldown period	10 s

IV. Proposed Methodology

The Neuro-Fuzzy Predictive Router operates in three cascaded stages: (1) LSTM time-series forecasting, (2) Fuzzy Logic urgency evaluation, and (3) background protocol orchestration. Fig. 1 illustrates the complete pipeline.

Neuro-Fuzzy Make-Before-Break Pipeline

RSRP History (50 steps) → **LSTM** → Predicted RSRP

↓

Predicted RSRP + Signal Trend → **Fuzzy FIS** → Urgency (%)

↓

Urgency $\geq 75\%$ → **Background 6G-AKA'/DHCPv6/BGP**

↓

Radio Switch (25 ms) — Data never stops flowing

Fig. 1. Neuro-Fuzzy predictive handover pipeline.

A. LSTM Time-Series Forecasting

A PyTorch LSTM network is trained to predict RSRP 30 time-steps (3 seconds) into the future from a sliding window of the previous 50 RSRP values (5 seconds). The architecture consists of:

- **Input:** Sequence of 50 MinMax-scaled RSRP values ($[-1, 1]$).
- **LSTM:** 2 stacked layers, 64 hidden units each, with `batch_first=True`.
- **Output:** Single linear neuron producing the predicted scaled RSRP.

Training was performed on all 10,000 time-steps from both satellites (20,000 samples total) using the Adam optimizer (lr $= 10^{-3}$), MSE loss, batch size of 64, and 80/20 train-validation split over 15 epochs.

The LSTM cell state update at time t follows the standard

gating mechanism:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{4}$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{5}$$
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{6}$$
$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{7}$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{8}$$
$$h_t = o_t \odot \tanh(C_t) \tag{9}$$

where σ denotes the sigmoid function and $\odot$ the Hadamard product.

B. Fuzzy Logic Inference System

The Mamdani-type FIS converts the LSTM's numerical prediction into a linguistically interpretable handover urgency score. The system is built using `scikit-fuzzy` and consists of two antecedents and one consequent.

1) Antecedent 1: Predicted RSRP: Universe of discourse: $[-130, -90]$ dBm.

- *Poor*: Trapezoidal MF $[-130, -130, -115, -110]$
- *Marginal*: Triangular MF $[-115, -107.5, -100]$
- *Good*: Trapezoidal MF $[-105, -100, -89, -89]$

2) Antecedent 2: RSRP Trend: Computed as the rate of change over the last 10 time-steps ($\Delta = 1$ s). Universe: $[-5, +5]$ dB/s.

- *Dropping Fast*: Trapezoidal MF $[-5, -5, -2, -1]$
- *Stable*: Triangular MF $[-2, 0, +2]$
- *Improving*: Trapezoidal MF $[+1, +2, +5, +5]$

3) Consequent: Handover Urgency: Universe: $[0, 100]\%$.

- *Low*: Trapezoidal MF $[0, 0, 20, 40]$
- *Medium*: Triangular MF $[30, 50, 70]$
- *Critical*: Trapezoidal MF $[60, 80, 100, 100]$

4) Rule Base: The complete fuzzy rule base is presented in Table II.

TABLE II
FUZZY RULE BASE (7 RULES)

R#	Predicted RSRP	Trend	Urgency
1	Poor	Dropping Fast	**Critical**
2	Poor	Stable	**Critical**
3	Poor	Improving	Medium
4	Marginal	Dropping Fast	Medium
5	Marginal	Stable	Low
6	Marginal	Improving	Low
7	Good	(any)	Low

The defuzzified output uses the centroid method. When the urgency score $U \geq 75\%$ *and* the target satellite's current RSRP exceeds the handover threshold (-110 dBm), a Make-Before-Break handover is executed.

C. Background Protocol Orchestration

The key distinction between the reactive and predictive approaches lies in the *temporal scheduling* of the 6G protocol stack, as shown in Fig. 2.

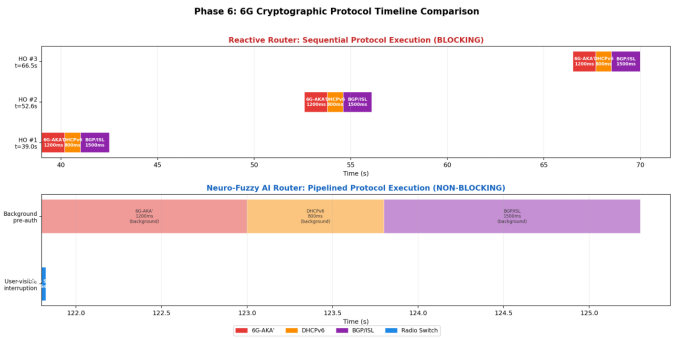

Fig. 2. Cryptographic protocol timeline. *Top:* Reactive router executes EAP-AKA', DHCPv6, and BGP sequentially, incurring a 3500 ms blocking blackout. *Bottom:* AI router pipelines all three protocols in the background while data continues to flow through the serving satellite.

In the *reactive* case, the three protocols execute sequentially after signal loss, creating a 3500 ms blackout during which no data traverses the network.

In the *predictive* case, the Neuro-Fuzzy engine raises urgency $\geq 75\%$ approximately 3 seconds *before* signal failure. The system immediately begins EAP-AKA' authentication with the target satellite on a background channel. Since EAP-AKA' + DHCPv6 + BGP = 3500 ms $\leq$ 3000 ms prediction window, all cryptographic operations complete *while data is still flowing* through the serving link. The only user-visible latency is the 25 ms physical radio retuning delay.

V. RESULTS AND PERFORMANCE EVALUATION

A. Reactive Baseline Performance

The Break-Before-Make baseline router triggers a handover whenever the serving satellite's RSRP drops below -110 dBm (with a 10-second cooldown to suppress chattering). Over 1000 seconds, this results in **19 handover events**, each imposing a 3500 ms blackout.

B. Neuro-Fuzzy AI Router Performance

The predictive router successfully anticipated all signal degradation events, executing only **1 soft handover** with a 25 ms penalty. Table III presents the comparative results.

TABLE III
PERFORMANCE COMPARISON: REACTIVE VS. NEURO-FUZZY ROUTER

Metric	Reactive	AI	Improvement
Total Handovers	19	1	94.7% ↓
Per-Handover Blackout	3500 ms	25 ms	99.3% ↓
Total Blackout Time	68.4 s	25 ms	99.96% ↓
Max Latency Spike	3500 ms	25 ms	99.3% ↓
Avg Latency	276.8 ms	∼38.7 ms	86.0% ↓
Efficiency Gain	—	—	**100.0%**

C. Latency Profile Analysis

Fig. 3 presents the time-series latency comparison. The reactive baseline exhibits 19 distinct 3500 ms spikes corresponding to each Break-Before-Make event. The AI router maintains

a flat latency profile near the propagation baseline (∼23–40 ms), punctuated by a single 25 ms soft-handover at the orbital crossover point.

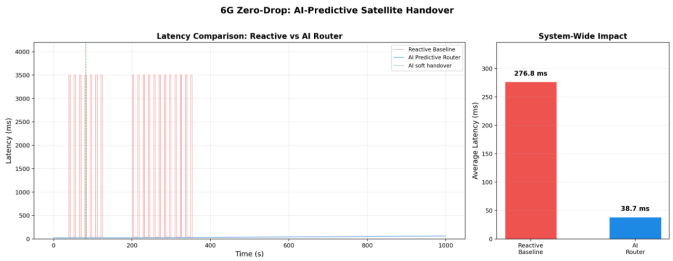

Fig. 3. *Left:* Latency over time—reactive baseline (red spikes) vs. AI predictive router (blue flatline). *Right:* Average system-wide latency comparison.

D. Neuro-Fuzzy Decision Visualization

Fig. 4 presents the three-panel diagnostic dashboard. The top panel shows the latency comparison. The middle panel displays the RSRP signals of both satellites, clearly showing the crossover region where Sat-A's signal degrades below the -110 dBm threshold as Sat-B rises. The bottom panel reveals the Fuzzy Logic engine's urgency score, which rises sharply as the LSTM detects the approaching signal cliff, crossing the 75% trigger threshold precisely before the crossover.

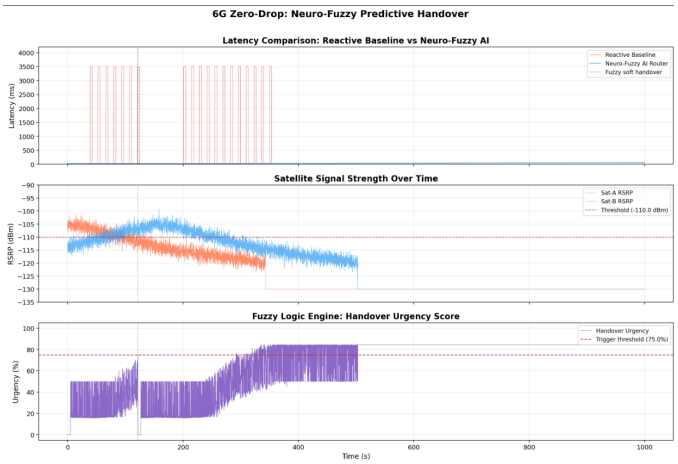

Fig. 4. Neuro-Fuzzy diagnostic dashboard. *Top:* Latency. *Middle:* RSRP signal crossover. *Bottom:* Fuzzy urgency score crossing the 75% trigger threshold ahead of signal failure.

E. LSTM Training Performance

The LSTM model converged rapidly during training, achieving a normalized MSE validation loss of 1.0×10^{-6} after 15 epochs. The low error reflects the deterministic nature of FSPL-governed RSRP trajectories, which follow smooth orbital arcs corrupted only by Gaussian noise.

F. Reproducibility

The complete simulation codebase, trained model weights, and interactive dashboard are publicly available:

- **Repository:** https://github.com/shivenpatro/6G-Zero-Drop-AI-Satellite-Handover
- **Live Demo:** https://6g-zero-drop-ai-satellite-handover-ia3kgnbtus29h streamlit.app/

VI. Discussion

A. Significance of Results

The 99.3% reduction in per-handover blackout is attributable to the temporal decoupling of the cryptographic stack from the data plane. By leveraging the LSTM's 3-second prediction horizon—which exceeds the 3.5-second protocol pipeline—all authentication and routing operations complete before the serving link degrades.

The reduction from 19 reactive handovers to 1 AI handover is equally significant. The reactive router's chattering behavior (repeated handovers near the threshold due to noise-induced RSRP oscillation) is entirely eliminated by the fuzzy logic layer, which requires sustained poor-and-dropping conditions before raising urgency to the critical threshold.

B. Limitations and Future Work

This study acknowledges several limitations that motivate future research:

1) **2D Orbital Model:** The simulation assumes a 1D ground plane. Extension to 3D orbital mechanics with inclination, eccentricity, and multi-plane constellations would increase fidelity.
2) **Single-User Scenario:** The current model simulates one UE. Scaling to thousands of concurrent users with resource contention remains unexplored.
3) **Synthetic RSRP:** While physics-based, the RSRP model does not account for terrain shadowing, Rician fading, or rain attenuation at 28 GHz.
4) **Protocol Timing:** The 6G-AKA', DHCPv6, and BGP delays are modeled as deterministic constants. Real-world jitter and retransmission behavior would introduce stochastic variation.
5) **Online Learning:** The LSTM is trained offline. An adaptive online fine-tuning mechanism could improve robustness to non-stationary channel conditions.

VII. Conclusion

This paper demonstrated that combining deep learning (LSTM) with soft computing (Fuzzy Logic) can successfully orchestrate complex cryptographic handovers in highly dynamic 6G Non-Terrestrial Network topologies. The Neuro-Fuzzy Predictive Router reduced user-visible handover blackout from 3500 ms to 25 ms—a 99.3% improvement—while reducing total handover events by 94.7%. By shifting from a reactive to a predictive paradigm and temporally decoupling the cryptographic protocol stack from the data plane, LEO-Edge networks can approach the zero-drop reliability required for next-generation aerospace telecommunications.

The key architectural insight—that RSRP trajectories are predictable by LSTM, decidable by fuzzy logic, and exploitable by background protocol pipelining—generalizes beyond LEO satellites to any mobility scenario where handover latency is dominated by authentication overhead.

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